

Integrate SBPA Workflow into RAP Business Object Application

SAP Build Process Automation — ABAP Environment Workflow Integration

Beginner

25 min

Customer/Partner License

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Overview & Objectives

In this final tutorial of the series, you will extend the RAP (RESTful Application Programming) Travel Business Object application to trigger the deployed **SAP Build Process Automation (SBPA)** workflow process directly from backend action logic.

Prerequisites

- Created and configured the RAP Travel application in ADT.
- Created, released, and deployed the workflow project in SAP Build Process Automation.
- Created the callback handler class `ZCL_RAP110_WORKFLOW_HANDLER_###`.

Step 1: Declare Custom Action in Behavior Definition

Open the Behavior Definition `ZRAP110_I_TRAVEL_###` in ADT and add a custom instance action to initiate the approval workflow:

```
// Add inside the Travel entity definition action ( features : instance )
createApprovalWorkflow result [1] $self;
```

Step 2: Implement Workflow Trigger Action in Behavior Implementation Class

In the Local Types class implementation (`BP_ZRAP110_I_TRAVEL_###`), add logic to call the SBPA REST API using the Destination service and HTTP client APIs:

```
METHOD createApprovalWorkflow. DATA: lv_payload TYPE string. " 1. Read Travel instance data
via EML READ ENTITIES OF zrap110_i_travel_### IN LOCAL MODE ENTITY Travel ALL FIELDS WITH
CORRESPONDING #( keys ) RESULT DATA(lt_travel). " 2. Construct JSON Payload for SBPA Process
Trigger " Payload contains Process Definition ID and Context Data " 3. Call SBPA REST API
using CL_HTTP_CLIENT / BTP Destination Service " Set overall_status = 'O' (In Approval) " 4.
Update Travel BO Instance status via EML MODIFY ENTITIES OF zrap110_i_travel_### IN LOCAL MODE
ENTITY Travel UPDATE FIELDS ( OverallStatus ) WITH VALUE #( FOR travel IN lt_travel ( %tky =
travel-%tky OverallStatus = 'O' ) ). result = VALUE #( FOR travel IN lt_travel ( %tky =
travel-%tky %param = travel ) ). ENDMETHOD.
```

Step 3: Expose Action in Behavior Definition & UI Metadata

1. Expose the action in the Projection Behavior Definition (`ZRAP110_C_TRAVEL_###`):

```
use action createApprovalWorkflow;
```

2. Add UI annotation in CDS Metadata Extension (`ZRAP110_C_TRAVEL_###.mext`) to render a button on the Fiori Elements UI:

```
@UI.lineItem: [{ position: 10, type: #FOR_ACTION, dataAction: 'createApprovalWorkflow',
label: 'Send for Approval' }] @UI.identification: [{ position: 10, type: #FOR_ACTION,
dataAction: 'createApprovalWorkflow', label: 'Send for Approval' }]
```

3. Save and activate all modified repository objects (`Ctrl + Shift + F3`).

Step 4: End-to-End Execution and Testing

1. Launch the SAP Fiori Elements Travel App preview from ADT.
2. Select a Travel entry with status *Open* ('O' or 'N') and click **Send for Approval**.
3. Verify in **SAP Build Process Automation - Monitoring** that a new process instance has been created.
4. Open ****My Inbox**** in SBPA/BTP to review, approve, or reject the approval task form.
5. Upon task completion, verify that the callback handler class updates the RAP Travel BO status to *Accepted* or *Rejected* in the backend database.

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